

Prediction of the temperature profiles using the generative model and the stacking ensemble method in a marginal sea

Kwangwoog Jung^a, Yang-Ki Cho^{a,b}, Myeong-Taek Kwak^{c,*}

^a School of Earth and Environmental Science, Seoul National University, Seoul, South Korea

^b Research Institute of Oceanography, Seoul National University, Seoul, South Korea

^c Department of Oceanography, Republic of Korea Naval Academy, Changwon, South Korea

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ABSTRACT

Various machine learning and neural network models using remote sensing data are useful for understanding and predicting the ocean status. Most machine learning methods have focused on predicting sea surface status because satellite data mainly provide only sea surface information. Subsurface data have been sparsely observed owing to difficulty accessing the ocean subsurface. Machine learning could be a viable option for predicting the subsurface status. Sea temperature profiles have been predicted using various combinations of machine learning methods and ensemble models. However, ocean observation data are insufficient for machine learning to resolve the complex spatiotemporal variability of sea temperature in marginal seas. We propose data augmentation of observation datasets to overcome sparse subsurface data in marginal seas. In the proposed method, remote sensing data are integrated with an augmented dataset to predict the temperature profile based on the stacking ensemble method. Further, we developed a neural network generative model-based solution to augment the observational dataset and make the preparation of datasets for training easier. The performance of the prediction model was evaluated using the observed dataset. The prediction model's performance was evaluated using the observed dataset. With artificially augmented data, the model showed improved accuracy, reducing RMSE from 2.51 to 1.20 in February, from 3.60 to 2.45 in August, demonstrating enhanced predictive capability and generalization. This paper is based on author's dissertation carried out at the SNU (Jung, 2022).

1. Introduction

The development of satellite and sensor technology enables us to easily obtain sea surface information, but there are limitations to directly obtaining subsurface information. Subsurface data are sparse, whereas many satellites routinely collect sea surface temperature (SST) and sea surface height (SSH). Recently, many researchers have attempted to estimate the ocean's vertical temperature profile using statistical or machine learning techniques (Jiang et al., 2017; Wang et al., 2021). Some studies have been conducted to estimate the subsurface temperature distribution of sea water vertically using satellite and Argo Float data (Han et al., 2019). Argo floats also have limitations in the precision or measurement of specific areas, such as near coasts and marginal seas, because the floats are not uniformly distributed (Roemmich et al., 2019). Recent work by Dubey and Vidyarthi (2025) demonstrated that coupling ground-observed meteorological data with CMIP6 simulations and machine-learning models (MVL, ANN, and

SVM) can substantially enhance rainfall forecasting accuracy. Their study underscores the importance of hybrid data-driven frameworks for regional climate prediction and an approach consistent with our proposed augmentation-ensemble design for marginal seas.

1.1. Background and motivation

The ocean stores approximately 93% of the world's energy, with subsurface temperature distribution playing a critical role in heat redistribution and climate regulation (Wang et al., 2021). Accurate knowledge of subsurface temperature profiles is essential for understanding ocean circulation, predicting climate variability, supporting fisheries management, and enabling naval operations (Schmidt et al., 2019). However, direct subsurface observations remain sparse and expensive compared to satellite-based surface measurements.

* Corresponding author.

E-mail address: mtkwak81@gmail.com (M.-T. Kwak).

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1.2. Previous studies and limitations

Machine learning techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have recently been used to predict the vertical temperature profile using surface information such as SST and SSH (Han et al., 2019). Prior studies have made significant contributions, but there are some limitations. The depth of data collection may limit the predicted value at a given depth. The measured locations in Argo-float were sparse and not fixed (Roemmich et al., 2019). While stationary data collection provides uniform datasets at fixed locations, such datasets are often insufficient in scale and variability. Therefore, augmentation of observational data is necessary to prepare sufficiently diverse datasets for effective model training and prediction of ocean physical properties and creating a model suitable for predicting ocean physical properties requires the preparation of sufficient datasets for model training. Although machine learning-based prediction models perform well in open oceans such as the Pacific Ocean, they are used on a limited basis in marginal seas because of the large spatiotemporal variability in the temperature and currents.

We summarize limitations as follows in brief.

Data scarcity: Machine learning models require large training datasets to capture complex patterns. In marginal seas such as the East/Japan Sea (EJS), the Mediterranean Sea, and coastal regions, observational data are often limited to a few monitoring stations with irregular sampling intervals. This sparse data distribution prevents models from learning the full range of oceanographic variability.

Spatiotemporal complexity: Marginal seas exhibit stronger meso-scale variability than open oceans due to boundary currents, eddies, river discharge, and complex topography (Seo et al., 2014). The EJS, for example, features the East Korean Warm Current (EKWC) and Ulleung Warm Eddy (UWE) with seasonal and interannual variability that create large temperature gradients over short distances. Models trained on open ocean data often fail to capture these localized dynamics.

Fixed observation locations: Unlike drifting Argo floats, coastal monitoring stations provide time series at fixed locations, which have the advantage of temporal continuity but limited spatial coverage. Existing machine learning approaches have not fully exploited this temporal richness while addressing spatial sparsity.

1.3. Research objectives and approach

This study addresses the fundamental challenge of predicting subsurface temperature profiles in marginal seas, where the scarcity and irregular distribution of in-situ observations pose significant constraints on model accuracy. Specifically, we aim to investigate two research questions:

- (1) Can generative models produce physically consistent and spatially coherent synthetic subsurface temperature data from sparse coastal observations?
- (2) To what extent does data augmentation using such synthetic data improve the predictive skill of subsurface temperature models compared to those trained solely on observational datasets?

To explore these questions, we develop a novel two-stage modeling framework specifically designed for the complex and data-limited environments of marginal seas. The framework integrates generative modeling with machine learning-based prediction to enhance the representativeness and robustness of subsurface temperature estimates.

Stage 1 - Data Augmentation: We apply advanced generative models—Triplet-based Variational Autoencoder (TVAE), Conditional Tabular GAN (CTGAN), and CopulaGAN—to augment a sparse observational dataset (31 observed profile) from a fixed monitoring station in EJS. These models are specifically designed for tabular

data with non-Gaussian, multimodal distributions characteristic of ocean temperature profiles.

Stage 2 - Subsurface Temperature Prediction: We construct a stacking ensemble that combines three base learners (K-nearest neighbors regression, support vector regression, random forest regression) with linear regression as meta-learner. The ensemble is trained on the augmented dataset (~4000 synthetic profiles) to predict temperature at 14 vertical levels using satellite-derived SST and SSH as inputs.

Our approach differs from previous studies in three key aspects: (1) explicit focus on data-sparse marginal seas rather than open oceans, (2) use of state-of-the-art generative models designed for tabular data and (3) rigorous validation of synthetic data quality through statistical testing and physical consistency checks.

In this study, the data augmentation step is based on learning the empirical statistical distribution of the observed vertical temperature profiles at a fixed marginal-sea station using GAN-based generative models. These models flexibly represent the non-Gaussian, multimodal structure of the limited samples and generate additional realizations that, after simple physical screening (realistic temperature ranges and vertically stable profiles), are used to train the prediction models and improve subsurface temperature estimates in this data-limited setting.

1.4. Study significance

Unlike previous studies that directly applied machine learning models to ocean data, this work proposes an integrated framework designed for data-sparse marginal seas, where limited observations lead to unstable machine learning training and degraded prediction performance.

Since most machine learning methods require a sufficiently large training dataset, direct application under such conditions can lead to unstable or unreliable predictions. To address this limitation, we evaluate whether machine learning-based subsurface temperature prediction becomes feasible when the effective training dataset is expanded using generative data augmentation and model uncertainty is reduced through ensemble learning.

1.5. Paper organization

The remainder of this paper is organized as follows. Section 2 describes the study area, observational and satellite datasets, and data preprocessing procedures. Section 3 presents the methodology, including detailed descriptions of the generative models (Stage 1) and stacking ensemble prediction framework (Stage 2). Section 4 presents results on synthetic data quality and temperature prediction performance. Section 5 discusses the implications, limitations, and future directions. Section 6 concludes with key findings and recommendations.

2. Study area and data

2.1. Study area

The Tsushima Current (TC) supplies heat and salt to the East/Japan sea (EJS) (Preller and Hogan, 1998). The TC is divided into two branches: one along the Japanese coast and the other along the Korean coast (Fig. 1) (Cho and Kim, 2000). This flow along the Korean coast is called the East Korean Warm Current (EKWC) (Cho and Kim, 1996; Kim et al., 2018). The EKWC turns eastward around Ulleung Island, forming the Ulleung Warm Eddy (UWE) (Kang and Kang, 1990; Kim et al., 1991; Katoh, 1994). The UWE, with a diameter of approximately 150 km, is located in the Ulleung Basin (Fig. 1). The size and location of the UWE vary seasonally and interannually (Kang and Kang, 1990; Isoda and Saitoh, 1993; Choi et al., 2004). The UWE plays a key ecological role in supporting a significant phytoplankton biomass (Kim et al., 2012).

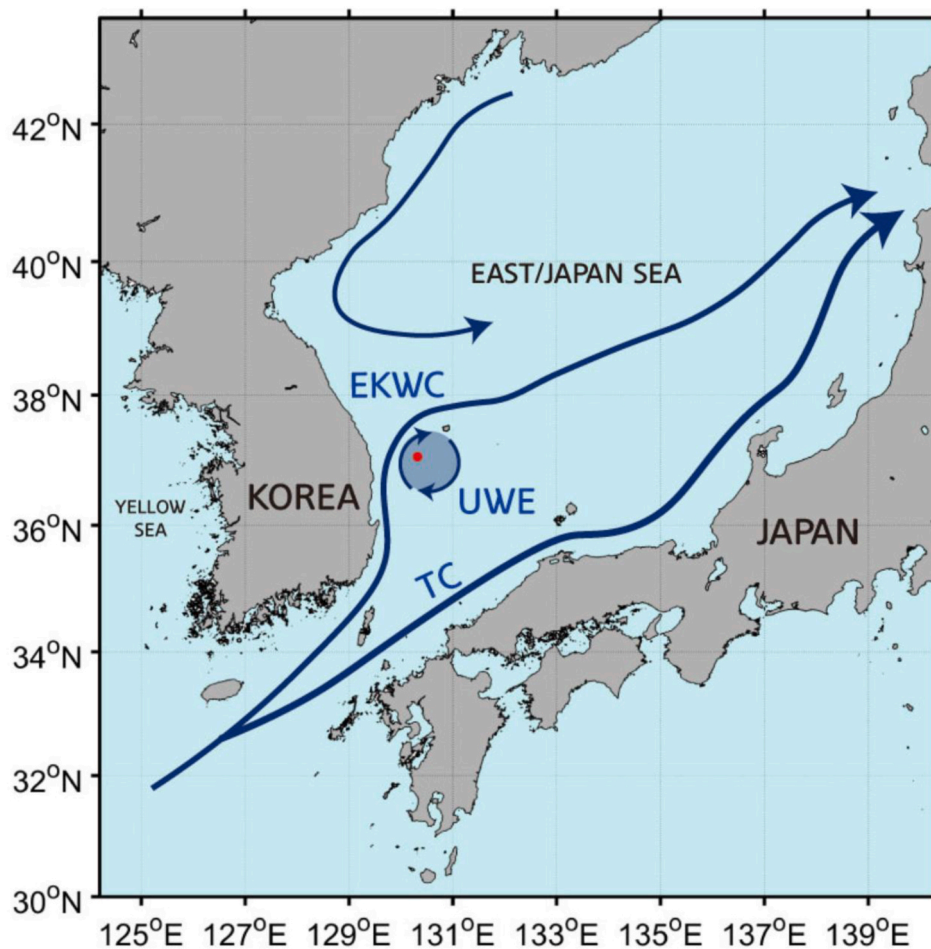


Fig. 1. Schematic currents in the study area and model domain. The red point (37.06°N, 130.31°E) represents a routine observation station selected for comparing the model and observation temperature profiles. TC, EKWC, and UWE denote Tsushima Current, East Korean Warm Current, and Ulleung Warm Eddy, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.2. Dataset

A station routinely observed by the National Institute of Fisheries Sciences (NIFS), which is located in the UWE, was selected for generating the sea subsurface temperature profile at our study point (37.06°N, 130.31°E, red circle in Fig. 1). The fluctuating characteristics of the temperature profile in the EJS render it suitable for testing the prediction performance of the proposed temperature profile model. Compared to the open sea, it is challenging to predict temperature profiles in marginal seas, such as the EJS, because many complex dynamic processes cause large variations in temperature and current (Seo et al., 2014; Joh et al., 2024).

The model has 14 vertical layers. The subsurface temperature profile was predicted using the NIFS's serial oceanographic observation dataset (NIFS, 2022). The SST dataset for the research domain was extracted from the advanced very high-resolution radiometer (AVHRR) instrument. The AVHRR has a spatial grid resolution of approximately 0.25°, and the temporal resolution is 1 day. These datasets were downloaded from the National Center for Environmental Information (NOAA, 2022). The sea surface temperature data among the NIFS observation datasets were also used as auxiliary data to prepare the SST data. We used the Copernicus marine environment monitoring service (CMEMS) gridded dataset for daily sea-level data (Copernicus, 2022). The horizontal resolution was 0.25°. The datasets are sea-level daily gridded data from satellite observations for the global ocean from 1993 to 2017. We used a dataset for the prediction model, which included absolute dynamic topography (ADT) and sea-level anomaly (SLA) from daily sea-level data. We

used data from 1993 to 2017 when both SST and SSH satellite data were available. The dataset was downloaded from the CMEMS climate data store (CMEMS, 2022). Training data were created using data from 1993 to 2012 as seed data. The model's performance was measured using test data from 2013 to 2017.

We created a dataset for GAN seed data by combining satellite data on the date when the measured temperature was present for each reference depth of the observation point. The data were removed without artificial interpolation when missing temperature values at the corresponding depths were found. Datasets were generated using only data when all observation data existed at the reference depth on the corresponding date. Initial dataset size was 37 and after preprocessed dataset was 31.

3. Methodology

In this study, we intended to create a machine learning model that predicts the subsurface temperature profile by combining satellite datasets such as SSH and SST with locally measured in situ temperature profile data. For this purpose, our methodological approach consists of two distinct stages. First stage is data augmentation stage; second stage is prediction stage. Fig. 2 shows the conceptual architecture for predicting the sea subsurface temperature profile.

3.1. Data augmentation stage

To augment the sparse temperature profile dataset, we experimented

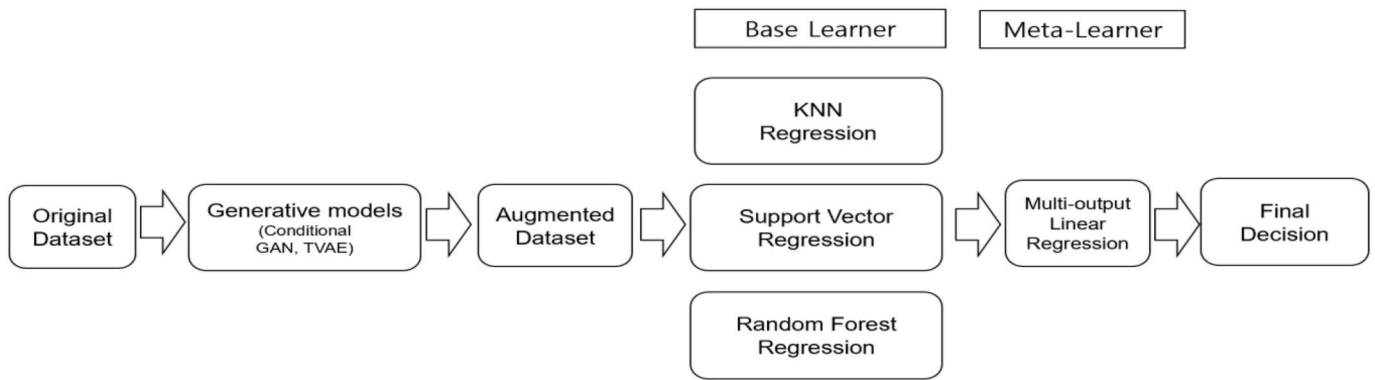


Fig. 2. Conceptual architecture for predicting subsurface temperature.

with various generative methods, including CTGAN, CopulaGAN, and TVAE (Xu et al., 2019). The observed datasets in Earth science are mainly tabular-type datasets, and continuous columns can have multiple modes. Some observed datasets may have non-Gaussian distributions, which are sometimes multimodal. Owing to these characteristics, there may be challenges in tabular data augmentation tasks using GANs (Xu et al., 2019). Much research has been conducted to overcome these challenges, and we applied the generative methods CTGAN, TVAE, and CopulaGAN in this study based on related research (Wang et al., 2024; Yadav et al., 2024). We summarized CTGAN and TVAE core comparative analysis and characteristics (Table 1).

In this study, we verified whether synthesized datasets based on generative models exhibited similar data distributions. To determine whether predictive models using various augmented data consistently performed better than models based on limited observational data, we utilized accuracy metrics such as RMSE and MAE on the selected individual models. The individual models selected in this study are representative KNN regression (Gou et al., 2018), support vector machine regression (Pourzangbar et al., 2023), and decision tree models (Gray et al., 2024), which are widely used in the marine science field. The results of this study are the prediction results obtained using an ensemble model that combines these individual models.

3.1.1. TVAE

Triplet-based variational autoencoders (TVAEs) are enhanced types of variational autoencoders (Ishfaq et al., 2018) that can learn latent representations with more fine-grained information.

Fig. 3 shows an example of latent representation, which is a key feature of the input data. The key features of dogs and cats are their ears and eyes. The latent representation is the sum of the latent features. The autoencoder, the middle layer of this network, contains a simplified representation of the input data and can be used to reconstruct the output.

TVAEs can learn an interpretable latent representation that preserves the original dataset's semantic structure by incorporating triplet constraints into the learning process. In each training iteration, the input

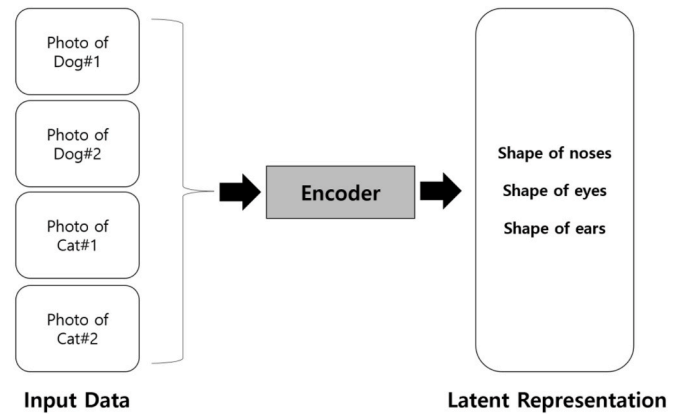


Fig. 3. Conceptual architecture of latent representation.

triplet is randomly sampled from the training dataset. Then, the triplet of images or data is inputted to the encoder network simultaneously to obtain their mean latent embedding (Ishfaq et al., 2018). A loss function over triplets to model the similarity structure over the image or data can be defined, as in Wang et al. (2014). Embedding, a method used to represent discrete variables as continuous vectors, is the process of converting high-dimensional data into low-dimensional data in the form of a vector such that the two are semantically similar (Jeevanandam, 2021).

3.1.2. Generative adversarial networks

The generative adversarial network (GAN) is a machine-learning method proposed by Goodfellow et al. (2014). A GAN consists of two networks: a generator and a discriminator. The core idea is that the generator is trained to generate fake or candidate data, and the discriminator is trained to distinguish between real and fake samples. The goal of training the generative network is to improve the discriminant network's error rate. In terms of data distribution, they compete with each other. A conceptual diagram is shown in Fig. 4.

Both networks have their own loss functions. The overall loss function of the GAN is given below; it is similar to the min-max problem (Goodfellow et al., 2014).

$$\min_{\mathcal{G}} \max_{\mathcal{D}} V(D, G) = \mathbb{E}_{x \sim P_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log (1 - D(G(z)))]$$

The definitions of the terms used are as follows.

Term	Definition
G	Generator model
D	Discriminator model
z	Random noise
x	Real data

(continued on next page)

Table 1

Core comparative analysis of CTGAN and TVAE.

Feature	CTGAN	TVAE
Best For	Large, complex datasets with non-linear relationships.	Smaller datasets where diversity and convergence are critical.
Training Speed	Slower; involves an adversarial game between generator and discriminator.	Faster; uses a straightforward Evidence Lower Bound (ELBO) loss for quicker convergence.
Fidelity/Accuracy	High for complex structures, though it may struggle with highly imbalanced data.	Often shows better statistical alignment (e.g., lower KL divergence) with real data distributions.

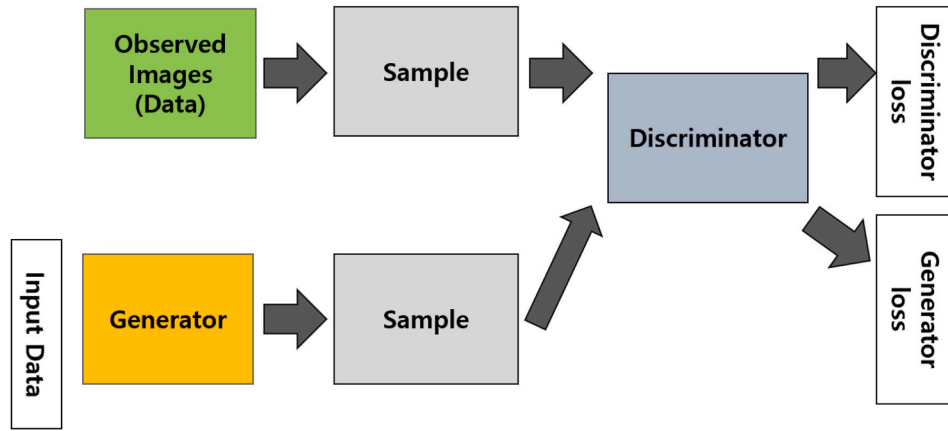


Fig. 4. Conceptual architecture of generative adversarial network.

(continued)

Term	Definition
$G(z)$	Data generated by Generator (synthetic data)
$p_{data}(x)$	Probability distribution of real data
$p_z(z)$	Probability distribution of synthetic data
$D(G(z))$	Discriminator's output when the generated data is an input
$D(x)$	Discriminator's output when the real data is an input

3.1.3. Conditional GAN

If we can determine the type of data to be generated through a GAN, GANs can be used for many scientific applications. If we suppose that both the generator and discriminator have some supplementary or auxiliary information y , GANs can be extended to a conditional model. Furthermore, y could be various types of supplementary information, such as class labels or different types of data. We can perform conditioning by inputting y into both the generator and discriminator as an additional input layer. The joint hidden representation in the generator combines the prior input noise $P_z(z)$ and y , and the adversarial training framework allows considerable flexibility in how this hidden representation is composed. In the discriminator, x and y are presented as inputs to a discriminative function (embodied again by a multilayer perceptron

in this case (Mirza and Osindero, 2014). Fig. 5 shows the conceptual structure of a basic conditional adversarial network. The generator synthesizes a fake sample ($G(z, y) = x^*|y$) using a random noise vector z and label y . Given the label, the fake sample's goal is to resemble the real sample as closely as possible. The discriminator takes a real sample and a label (x, y) , as well as a fake sample and the label used to generate it ($x^*|y, y$) (Langr and Bok, 2019). The discriminator learns to distinguish between real data and matching pairs from real sample-label pairs, and how to identify fake data-label pairs from a generator's sample. The discriminator outputs a single probability that the input pair is real data, and computes it using the activation function sigma of the sigmoid.

CTGAN is a GAN-based method for generating tabular data using the data distribution of a tabular sample dataset (Xu et al., 2019). CopulaGAN is a CTGAN model variant that uses a cumulative distribution function-based transformation (Synthetic Data Vault (SDV), 2022). The dataset of the sub-surface temperature profile is generally tabular data, and the temperature at each depth is not linearly dependent on the observed depth. TVAE with a variational autoencoder has been used to generate datasets with high performance and flexibility (Xu et al., 2019). We can prepare training datasets for the prediction model and apply them to train the prediction models, which are then used for the ensemble model, using the proposed augmentation architectures based

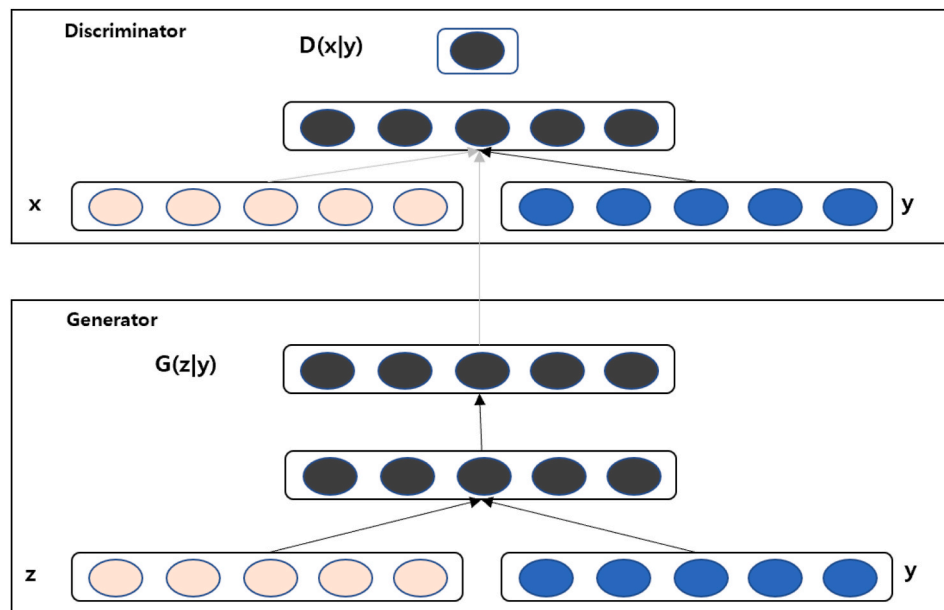


Fig. 5. Conceptual architecture of a conditional generative adversarial network.

on generative models. Enhancing model training is possible after artificially augmenting meaningful datasets.

3.1.4. Data augmentation model development

3.1.4.1. Configuration of the generative model. We tested several generative approaches (TVAE, CTGAN, CopulaGAN) due to its ability to represent non-Gaussian and multimodal distributions typical of subsurface temperature profiles. The generator is trained separately for February and August so that it can explicitly account for the strong seasonal contrast in stratification.

The training dataset for the generative model consists of:

Vectorized temperature profiles at 14 depths (0–500 m) from the NIFS station,

A categorical variable indicating the target month (February or August).

CTGAN is trained using the standard adversarial learning framework, where a generator network learns to produce synthetic profiles and a discriminator network learns to distinguish them from real observations until convergence.

3.1.4.2. Validation and filtering of synthetic data. After training, we generate a large pool of synthetic profiles and then apply a series of statistical and physical checks:

Distributional similarity:

For each depth and season, observed and synthetic marginal distributions are compared using p-value test. Synthetic values outside the climatological temperature range of the EJS are discarded.

Regime coverage:

Key metrics such as winter mixed layer depth, summer thermocline depth and gradient, and UWE core temperature anomalies are compared between observed and synthetic subsets to confirm that the synthetic data appropriately populate these regimes.

The synthetic profiles that pass all checks are paired with corresponding surface predictor vectors (SST, SSH, and month, sampled from the empirical surface distributions for the same season). The final augmented training dataset is then constructed by combining observed and synthetic cases:

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t \in \text{obs}} \cup \{(\tilde{\mathbf{x}}_j, \tilde{\mathbf{y}}_j)\}_{j \in \text{syn}}.$$

This augmented dataset serves as the input for the prediction-model development described in the next section.

The models used in this study were developed in Python using the TensorFlow-based Keras library and PyTorch-based SDV libraries (SDV, 2022). Generative model codes (that is, some types of GANs and TVAE used to generate sample data) were deployed using Python Jupyter notebooks on AMD 10 cores and Nvidia RTX-3090. Augmented datasets were used for several base models to construct an ensemble model for the prediction of sea subsurface temperature profiles. Generally, the ensemble model is more accurate than the standalone model at predicting values. We designed and implemented ensemble stacking methods using several candidates to improve performance.

3.2. Prediction stage

In the prediction model stage, we chose the ensemble model method, which allows us to combine and test various models to measure the effectiveness of the synthetic data created in the first stage. In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance than any of the constituent learning algorithms alone (Zhang and Ma, 2012).

In the stacking regression for the ensemble, we chose the K-nearest neighbors regression (KNNR) model, support vector regression (SVR), and random forest regression (RFR) as base learners and the multioutput linear regression (LR) model as the meta-learner in our study (Kalule

et al., 2023). Every base learner generates the predicted values based on their own algorithm, and they are used as datasets for the meta-learner.

3.2.1. Stacking ensemble

In ensemble learning, three major methods aim to combine base models or weak learners. Bagging and boosting learn homogeneous base models and combine them using a deterministic strategy or process (Rocca, 2019). Stacking learns heterogeneous weak learners in parallel and combines them by training a metamodel to output a prediction based on different base model predictions. Ensemble stacking, or stacked generalization, involves training a learning algorithm to combine the predictions of several other learning algorithms (Brownlee, 2021; Kadkhodaei et al., 2020; Rocca, 2019). First, all other algorithms are trained using the available data. Then, the combiner algorithm is trained to make a final prediction with all the prediction outputs of the other algorithms as additional inputs. Stacking typically outperforms any standalone trained model. Fig. 6 shows the conceptual architecture of the stacking ensemble. In this study, we used the ensemble stacking method to combine weak learners that stand out in individual models to build a model with better performance. Regression models, which are traditionally used to estimate numerical values, such as KNNR, SVR, and RFR, were chosen as base models.

Furthermore, we constructed a metamodel using a multioutput linear regression model that performs well in multiple predictions. Multioutput regression is a regression problem that involves predicting two or more numerical values given an input example (Brownlee, 2021). In this study, we employed a multioutput regressor to predict multiple subsurface temperatures by depth using SSH and surface temperature.

3.2.1.1. K-nearest neighbors regression. The k-nearest neighbor (k-NN) algorithm is a non-parametric supervised learning method used for classification and regression (Atteia et al., 2021). The K-NN regression output is the property value for the object, and the value is the average of the values of the k-nearest neighbors. K-NN estimates the association between the input and response variables using feature similarity (Yao and Ruzzo, 2006). In the k-NN regression, the response variables are approximated by averaging the observations in the nearest neighborhood of the input instance using similarity measures (Ali et al., 2019).

Gou et al. (2018) used KNNR to refine the existing datasets for thermocline research, and Liu et al. (2019) used KNNR to develop a method for constructing high-resolution ocean models and found that the proposed KNNR model was used to refine seawater thermocline data and improve the data resolution on their vertical gradient.

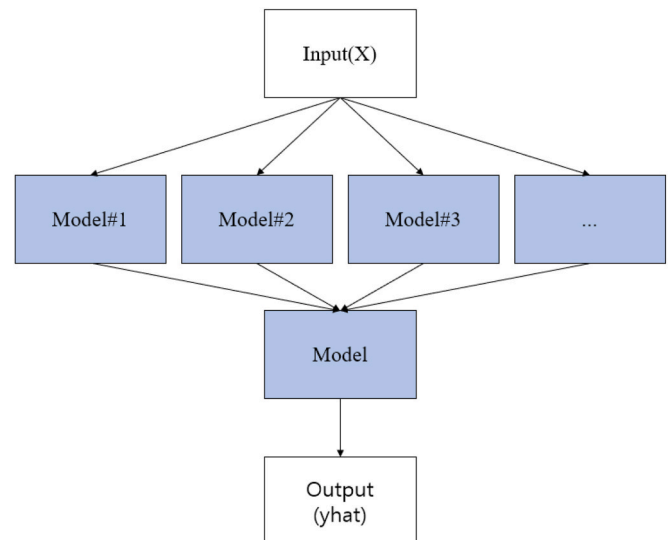


Fig. 6. Conceptual diagram of the stacking ensemble.

We chose KNNR as the base learner because of its approximation performance and the results obtained in recent thermocline research cases in which it was employed.

3.2.1.2. Support vector regression. SVR is an extended algorithm of the support vector machine (SVM), which is a classic and powerful machine learning algorithm for solving nonlinear regression problems (Brereton and Lloyd, 2010). SVR calculates the loss function based on structural risk minimization, allowing a deviation of ϵ between the model output and the real value. This differs from the traditional regression model, which is based on the error between the model output and the real output. This can avoid the disadvantages caused by pursuing experiential risk minimization. SVR is a model that uses high-dimensional feature spaces, but penalizes the resulting complexity using a penalty term augmented with the error function, making it suitable for fitting high-dimensional data with comparatively fewer samples (Balogun and Adebisi, 2021; Balogun et al., 2021). The basic idea of SVM is to map multi-dimensional data onto a higher-dimensional feature space. A hyperplane linearly separates the original data while maximizing the margin between different classes (Borges, 1998).

Using SVM, the sub-sea surface temperature anomaly in the Indian Ocean has been estimated from satellite measurements of sea surface parameters (with SSTA, SSHA, and SSSA as input attributes) (Su et al., 2015). Furthermore, Li et al. (2017) evaluated the performance of an SVM-complementary ensemble empirical mode decomposition model to estimate SST in the northeast Pacific Ocean. In another study, Jiang et al. (2018) evaluated the prediction performance of LR and SVR for SST in the Canadian Berkley Canyon. Water depth and coordinate information, such as latitude and longitude, were used as input variables. These input variables have seldom been used to assess SST in previous studies. The results showed that SVR provided estimates closer to the observed data than LR.

3.2.1.3. Random forest regression. In this study, RFR was chosen as the base learner to create the ensemble model. RFR creates robust estimates using an ensemble of decision trees, frequently without requiring data pre-processing, making it an effective “off the shelf” method (Louppe, 2014). Decision trees are useful for determining nonlinear relationships between the target variables and input features (Auret et al., 2012). Gregor et al. (2017) used SVR and RFR to estimate CO₂ levels in the Southern Ocean and achieved good prediction performance. A random forest is also a meta-estimator that fits several classifying decision trees on various subsamples of the dataset and uses averaging to improve the predictive accuracy and control overfitting (scikit-learn.org, 2022). We chose RFR for reasons such as nonlinear relationships and predictive accuracy.

3.2.1.4. Linear regression. Traditionally, linear regression analysis has been widely used in various Earth science fields. Linear models have been widely used in ocean prediction because they require minimal data input and are relatively simple. Although simple, it is effective in identifying trends and provides important insights for understanding and analyzing overall trends. It is widely used in ocean science to predict water temperature distributions and analyze trends. Many scientists have used linear regression models (Morrill et al., 2005; Krider et al., 2013) in ocean sciences. Feng et al. (2020) developed a multiple linear regression algorithm for sea surface temperature retrieval using one-dimensional synthetic-aperture microwave radiometry. The regression method is a strong candidate for determining the relationships among a variety of properties, such as sea surface temperature, sea surface height, and depth.

We also need to analyze the correlation between sea surface temperature and sea surface height and depth. Leuliette and Wahr (1999) studied coupled pattern analysis of sea surface temperature and TOPEX/Poseidon sea surface height. They showed that the spatial

correlation is strong in both the Atlantic and Pacific. The good temporal and spatial agreement between the SSH and SST fields suggests that a robust regression between fields may have some physical significance. With reference to the results of previous studies and the robustness of the model, we chose linear regression, one of the most common statistical methods, as an ensemble member in many oceanic analyses.

3.2.2. Accuracy

In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance.

For the performance evaluation of the machine learning models, we utilized the commonly used metrics mean absolute error (MAE) and root mean square error (RMSE) (Chai and Draxler, 2014; Vidyarthi et al., 2020). RMSE evaluates the residual between the observed and predicted values and is particularly sensitive to large errors. MAE is less sensitive to extreme values than RMSE (Ait-Amir et al., 2015).

We also evaluated the correlation coefficient to ascertain how well the predicted data fit the predicted value. The model was trained on the synthetic dataset and we calculated the correlation coefficient between the predicted and observed values for the synthetic dataset.

3.2.3. Prediction model development

3.2.3.1. Input-output configuration. The prediction models are designed to estimate the subsurface temperature profile at 14 standard depths from surface variables. The input vector $\mathbf{x}_t$ includes:

- Sea surface temperature (SST),
- Sea surface height (SSH),
- Temporal indicators (month)

While the output vector $\mathbf{y}_t$ contains temperatures at 14 depths between 0 and 500 m. All predictors are standardized before training, and the problem is formulated as a multi-output regression task.

3.2.3.2. Base prediction models. On the augmented dataset $\mathcal{D}_{\text{train}}$, we develop three complementary base models, chosen to represent different classes of machine-learning regressors.

3.2.3.2.1. K-nearest neighbors regression (KNNR). KNNR serves as a non-parametric local analogue model. For a new input $\mathbf{x}_*$, the model identifies the K nearest neighbors in the standardized predictor space and predicts the temperature at each depth as the average of their observed values:

$$\hat{\mathbf{y}}_{\text{KNNR}}(\mathbf{x}_*) = \frac{1}{K} \sum_{i \in \mathcal{I}_K(\mathbf{x}_*)} \mathbf{y}_i.$$

This approach is particularly suitable for winter, when the water column is nearly homogeneous and similar surface conditions tend to correspond to similar vertical profiles.

3.2.3.2.2. Support vector regression (SVR). SVR is employed to capture nonlinear relationships between surface conditions and subsurface temperature, especially in the presence of strong thermoclines. An RBF kernel is used, and hyper-parameters (C , γ , ϵ) are tuned via cross-validation on the augmented dataset. An independent SVR model is trained for each depth, allowing flexible depth-wise responses to the same set of surface predictors.

3.2.3.2.3. Random forest regression (RFR). RFR is used to model nonlinear interactions and regime shifts that may be difficult for KNNR or SVR to capture. For each depth, an ensemble of regression trees is trained on bootstrap samples of the augmented dataset, with random feature selection at each split. The number of trees, maximum depth, and minimum samples per leaf are tuned to balance bias and variance. RFR is particularly useful near UWE edges and during seasonal transition periods, where vertical temperature structure can change abruptly.

3.2.3.3. Stacking ensemble. To exploit the complementary strengths of the three base models, a stacking ensemble is constructed:

Level-1 predictions:

For each input and depth, KNNR, SVR, and RFR produce separate temperature predictions.

Meta-learner:

These three predictions are then used as inputs to a multi-output linear regression meta-model, which learns depth-specific combination weights:

$$\hat{y}_{d,ens}(\mathbf{x}_t) = w_{1,d}\hat{y}_{d,KNNR}(\mathbf{x}_t) + w_{2,d}\hat{y}_{d,SVR}(\mathbf{x}_t) + w_{3,d}\hat{y}_{d,RFR}(\mathbf{x}_t) + b_d.$$

3.2.3.4. Extensibility of the prediction framework. The prediction-model development is intentionally designed to be extensible. While this study focuses on KNNR, SVR, and RFR as representative models, the same augmented dataset and stacking framework can be applied to other algorithms, such as multivariate linear regression or artificial neural networks. The main contribution is therefore the framework that couples physically validated data augmentation with a flexible ensemble of prediction models, rather than the superiority of any single algorithm.

3.2.4. Training and prediction processes

The quality of the training data is crucial in the model training step. The prediction model was trained with the synthesized data based on the generative models. Approximately 4000 units of synthesized data were produced based on the observed data from 1993 to 2012. Individual models were also trained with the synthesized dataset, and RMSE and MAE were evaluated. Each individual model was used as a base model to construct the final stacking ensemble model, which was also trained with the same synthesized dataset.

Multiple prediction models were applied to take advantage of their complementary strengths. KNNR is effective at capturing local patterns in the data, while SVR excels in modeling non-linear relationships. RFR

is robust to overfitting and can effectively handle complex interactions among features. By combining these models into a stacking ensemble, we aimed to leverage their individual strengths to enhance prediction accuracy. The linear regression model used as the meta-learner aggregates the predictions from the base models to produce a final output that is more reliable and generalizable. This ensemble approach is intended to improve the model's performance by reducing bias and variance compared to individual models.

The prediction model trained on the synthesized data was used for prediction. The test datasets for the prediction consisted of samples in February and August, respectively, from 2013 to 2017. The performance of the prediction model was evaluated by comparing the predicted values with the observed values and HYCOM reanalysis values (HYCOM Consortium, 2022).

4. Results

4.1. Data generation

High-quality datasets are critical for the prediction model's performance in machine-learning approaches. Real and observed datasets may be costly and challenging to measure and acquire. In this study, generative models were used to generate subsurface temperature datasets, which are difficult to obtain, and the similarity was determined by comparing the synthetic and observed data distributions (Figs. 7 and 8).

Figs. 7 and 8 are plots of the histogram and kernel density estimation (KDE) together. Comparing the density estimation method using histograms with the KDE methods, the histogram method is discrete and increases the binary value corresponding to each dataset, resulting in discontinuity. The KDE method has the advantage of providing a smooth probability density function (PDF). In other words, the PDF obtained through KDE can also be seen as smoothing the histogram PDF, and the degree of smoothing depends on which bandwidth value kernel function

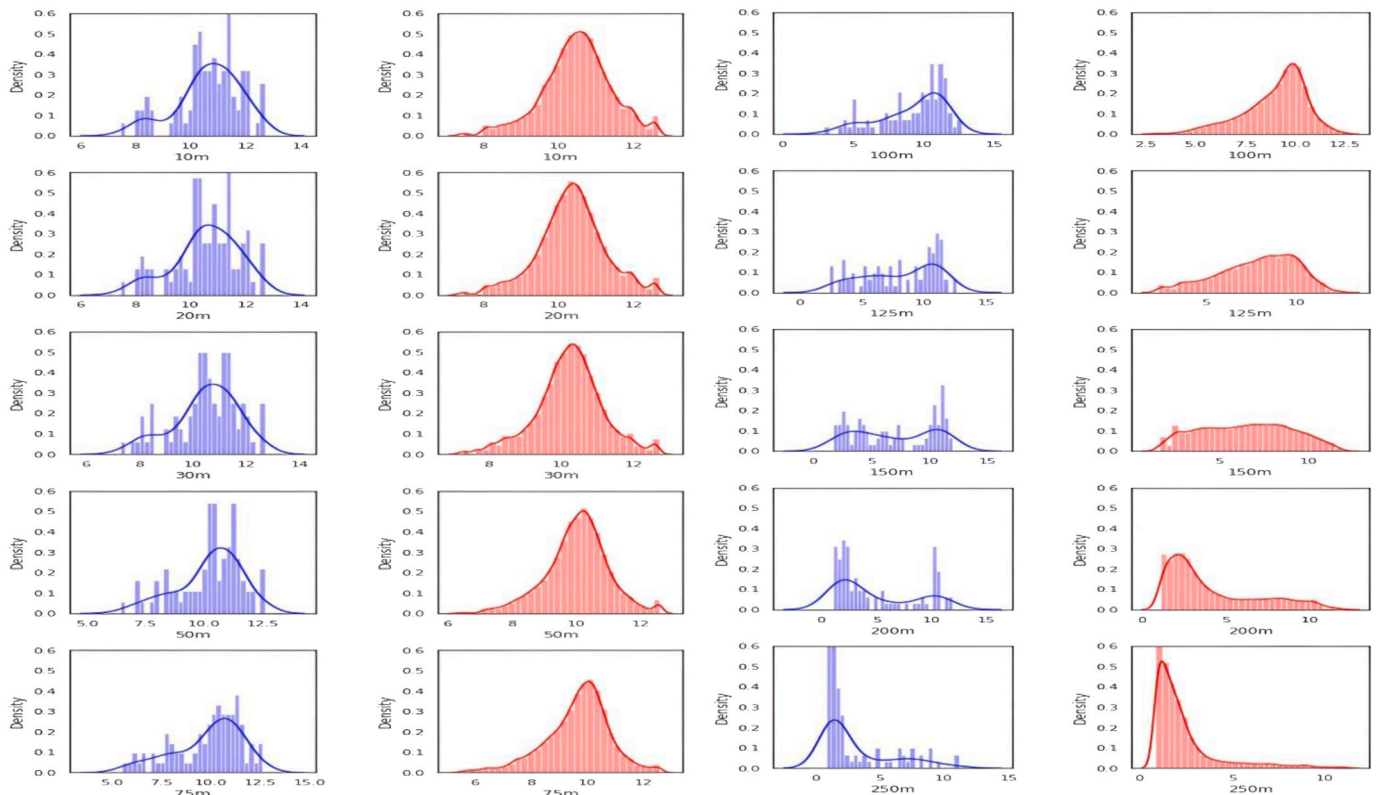


Fig. 7. Data distributions of observed (blue) and synthetic (red) datasets in FEB from 1993 to 2011. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

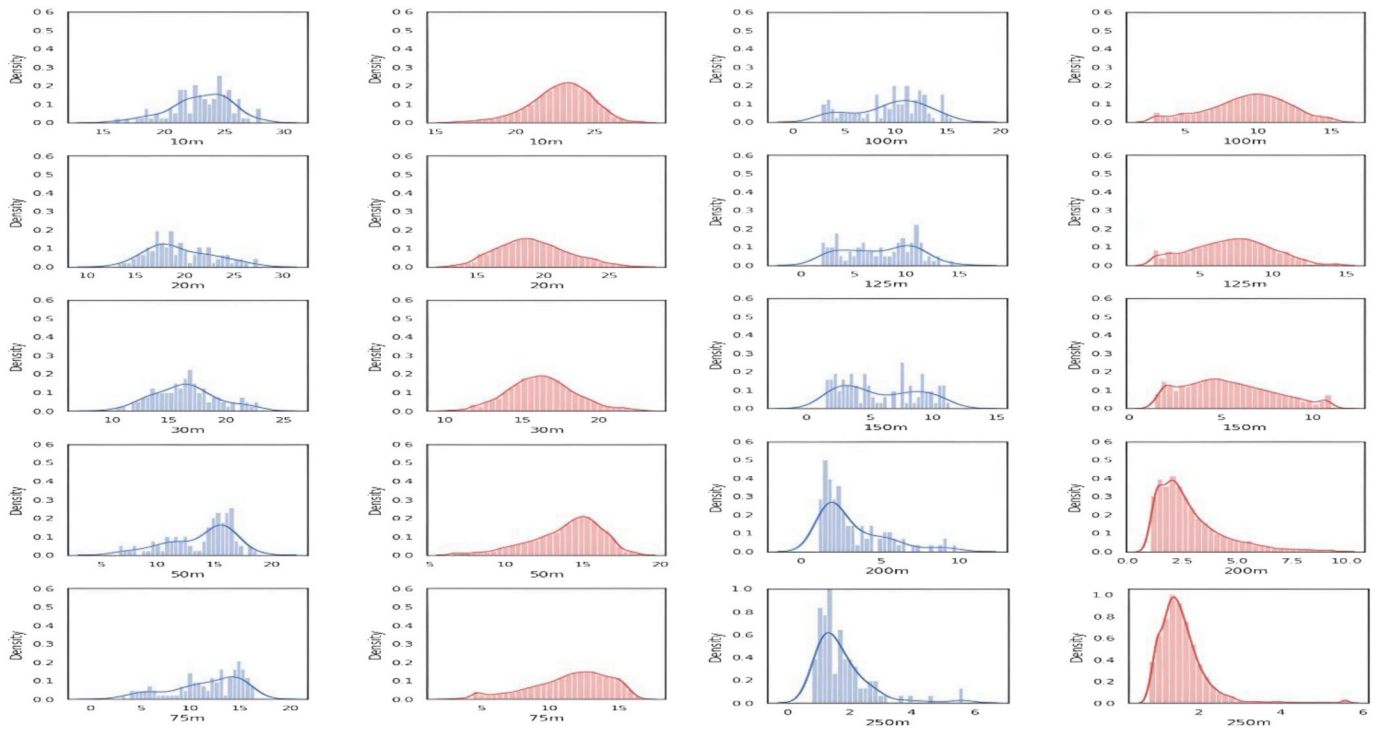


Fig. 8. Data distributions of observed (blue) and synthetic (red) datasets in AUG from 1993 to 2011. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

is used.

In this study, we attempted to generate subsurface profile data using TVAE and GANs and visualized the histogram and difference matrix of observed and synthetic datasets. Figs. 7 and 8 show the density histogram of the synthetic data according to depth, and these histograms provide information on the similarity between the synthetic and observed datasets. The difference matrix of the observed and synthetic datasets shows the similarities and differences between them (Figs. 9 and 10).

One of main purpose of this study is to generate more data reflecting the typical seasonal differences in the temperature profile of the EJS. A strong seasonal thermocline is developed in the EJS during the summer (Cho and Kim, 1998). The maximum gradient of thermocline reaches to 0.36 °C/m in August (Chu et al., 2001). The surface mixed layer is deepened due to increased wind-driven mixing during winter, resulting in disappearance of seasonal thermocline (Cho and Kim, 1998).

We evaluated synthesized datasets through generative model how accurately captures the seasonal variations in the temperature profile of

the EJS. Seasonal differences in temperature distribution are pronounced, particularly when considering the distinct vertical temperature profiles observed in summer and winter (Chu et al., 2001).

The shapes of the histogram between observed and synthetic data are similar, although the densities between both are somewhat different in February (Fig. 7). The synthetic data also shows a similar distribution of histogram with the observation (Fig. 8). The difference matrix indicates that the gap between the observed and synthetic data is small, and they are demonstrating their similarity (Figs. 9 and 10). The maximum differences are below 0.3 in February and August, respectively.

In this study, we compared the accuracy metrics of the candidate models, such as K-nearest neighborhood, SVR, random forest, and linear regression for selecting ensemble model members. Several models were used for the ensemble model, and we chose the base learners for the stacking ensemble based on previous studies and the values of the MAE and RMSE accuracy metrics. We chose an observation point close to the UWE and evaluated the MAE and RMSE of the base-learner models using the observed and synthetic datasets.

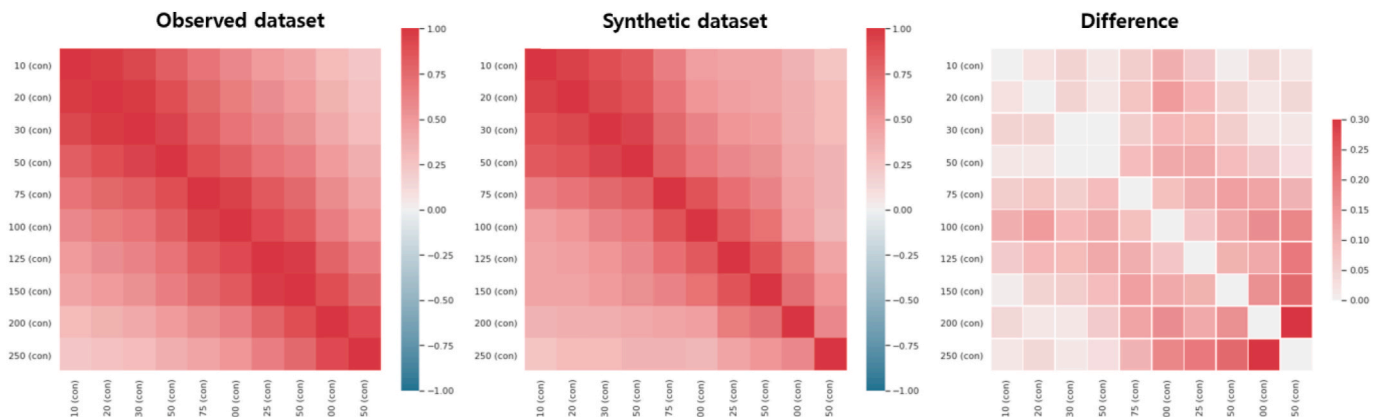


Fig. 9. Difference matrix of observed and synthetic datasets in FEB from 1993 to 2011.

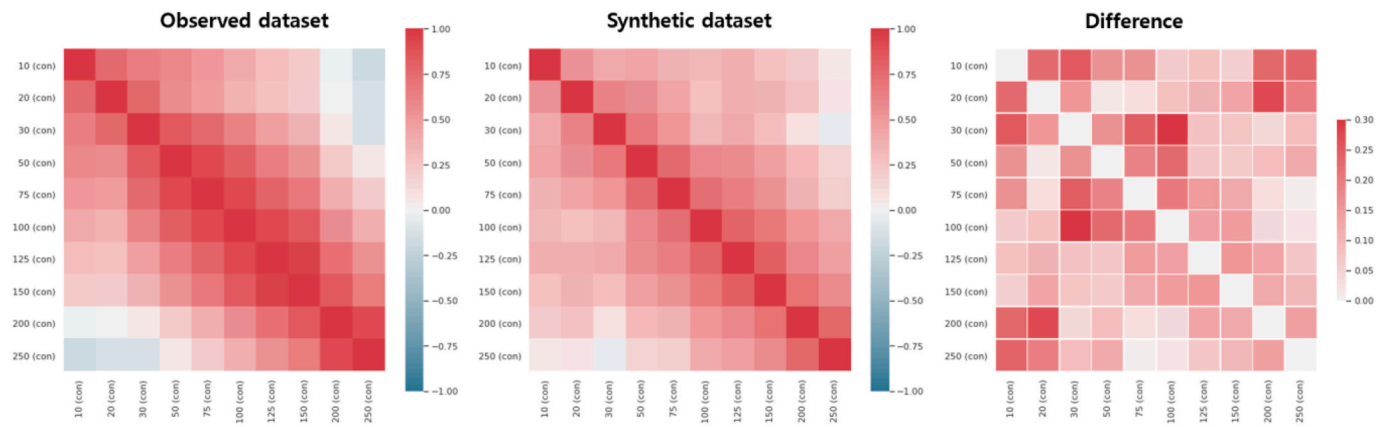


Fig. 10. Difference matrix of observed and synthetic datasets in AUG from 1993 to 2011.

The validation of generated dataset was performed by additional statistical analyses such as p-value test (Confidence level: 99%) for significance testing and distance calculation for quantitative similarity metrics.

The p-value test on the synthetic data sets shows that the real data set and the synthetic data are statistically equivalent. The synthetic data sets of FEB also show statistically equivalent except some depth (30 m, 200 m, 250 m) (Table 2).

For evaluation of quantitative similarity metrics, we calculate distance of synthetic data set and real data set. Fig. 11 showed distance between synthesized data set and real data set. The calculated distances are between 0.5 and 1.5 mainly.

In Fig. 12, the MAE and RMSE of the regression models KNNR, SVR, RFR, and LR using the observed and augmented datasets are shown. The MAEs and RMSEs of the models using the observed dataset were higher than those of the models using the synthetic dataset. This means that the accuracy of the prediction model using the synthetic dataset is better than that of the observed dataset in this study. When we compared the MAE and RMSE of the individual models to those of the stacking ensemble prediction in Table 3, the accuracy metrics of the stacking ensemble prediction were better.

4.2. Model prediction

Fig. 13 shows the prediction results of the stacking ensemble model. In this study, data were synthesized using data from a station in the UWE from 1993 to 2012, and used as training data for the model. Then, using the data for five years from 2013 to 2017 as test data, we measured the stacking ensemble's model prediction performance. Data from February for winter and August for summer were used to compare temperature profiles during seasonal changes. The in-situ observations used for validation were collected on specific survey days when weather and sea conditions allowed safe ship operations. Because hydrographic surveys in the Ulleung Warm Eddy area are conducted approximately once per

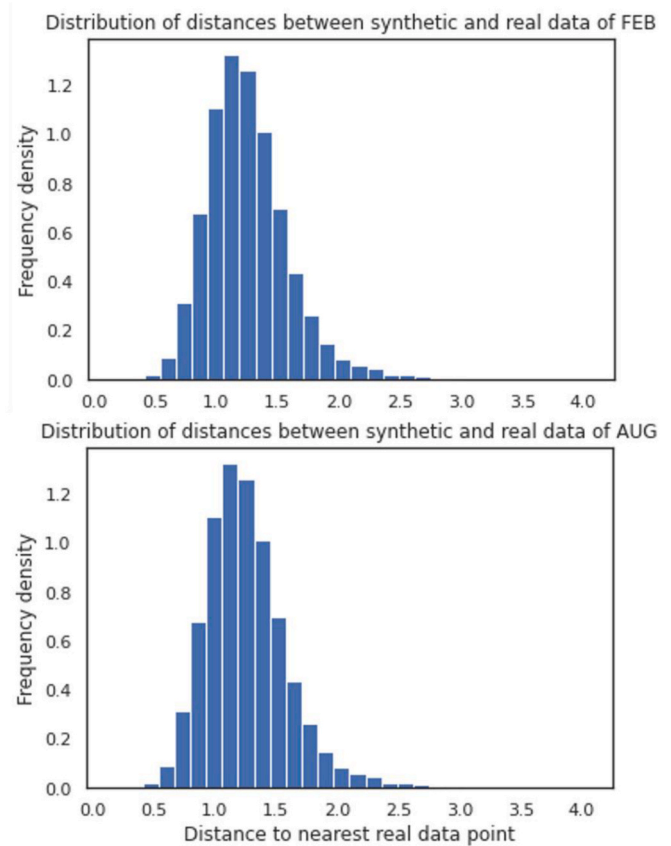


Fig. 11. Distribution of distances between synthetic and real data of FEB (upper) and AUG (lower).

Table 2
p-value test of synthesized dataset for statistical significance test.

Depth (m)	FEB	AUG
10	9.16393039e-02	0.48013835
20	2.40703686e-02	0.42295568
30	5.65114715e-03	0.74085054
50	2.05213731e-02	0.50115135
75	1.39884379e-01	0.74705859
100	1.58072448e-01	0.82429051
125	3.35103532e-01	0.4132323
150	7.30893413e-02	0.13186588
200	2.51290877e-03	0.02096151
250	1.52691924e-04	0.01578775

month, the observation dates are irregular and limited. Therefore, the profiles shown in Fig. 13 correspond to actual observation days. For clarity, February is presented as a representative month.

As shown in the observed temperature profiles, the UWE, which can be characterized by homogeneous water from the surface to about 200 m depth, appears in winter. However, the UWE mostly disappeared, and a strong thermocline appeared in summer. The model results accurately predicted the seasonal change in the temperature profile over the entire period, except for August 2015, when the remnant of the UWE was present in the subsurface.

To evaluate the prediction of the model visually, the HYCOM analysis data, the predicted value of our research model, and the actual observation values were displayed together. The ensemble model

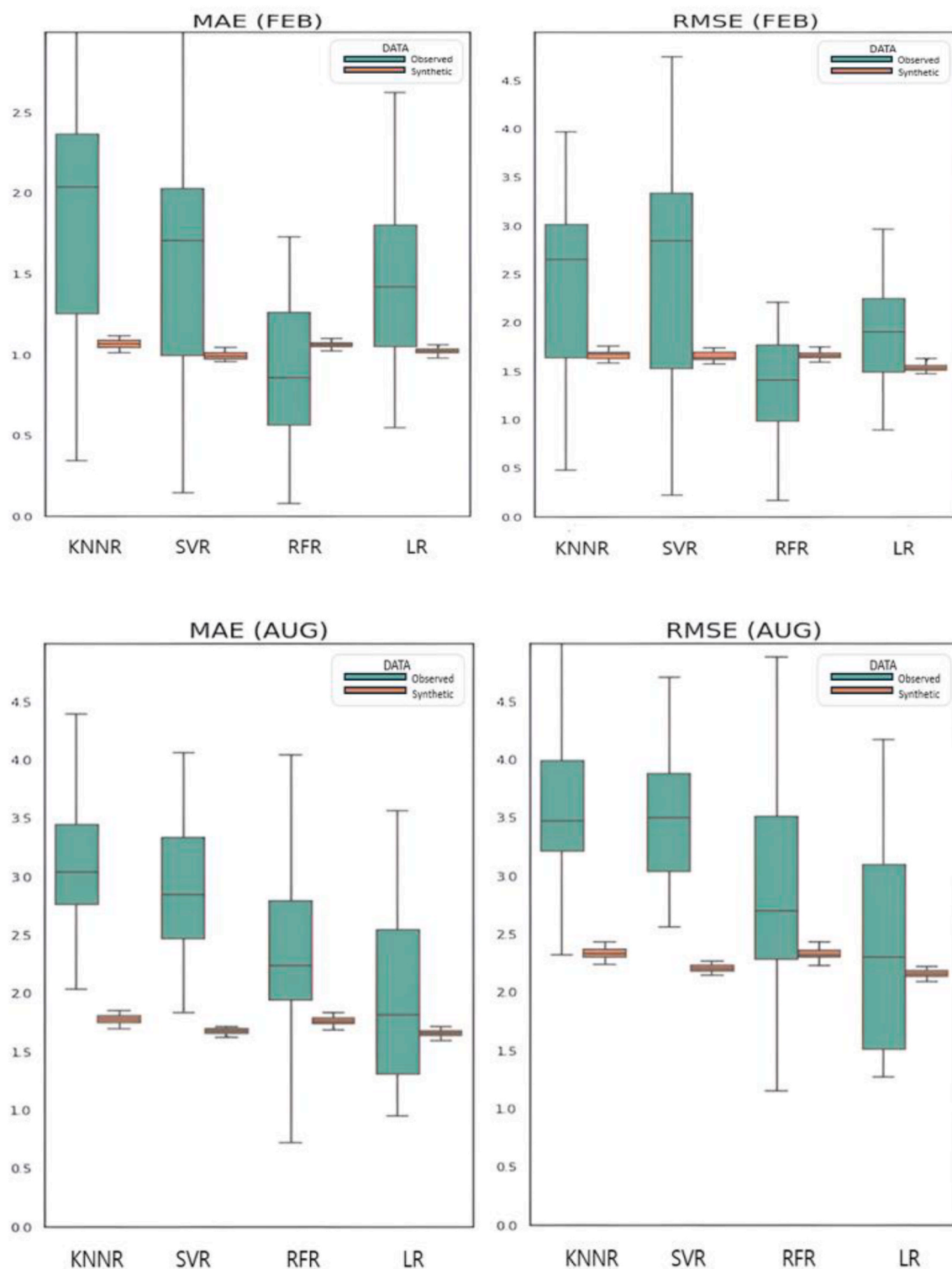


Fig. 12. MAE and RMSE of regression models (KNNR, SVR, RFR, LR) using FEB and AUG datasets.

Table 3

Model working efficiency and accuracy metrics on the synthetic dataset (FEB, AUG).

Dataset type	Synthetic Dataset (FEB)			Synthetic Dataset (AUG)		
	MAE	RMSE	r	MAE	RMSE	r
Accuracy	0.96	1.20	0.93	1.92	2.45	0.97

consistently achieved lower RMSE (1.20) MAE (0.96) compared to HYCOM's RMSE (3.72), MAE (1.48) in February.

4.3. Model working efficiency analysis

To evaluate the model working efficiency, we used the MAE and RMSE in this study. The MAE was 0.96 and 1.92 and The RMSE was 1.20 and 2.45 in February and August, respectively. The performance of the

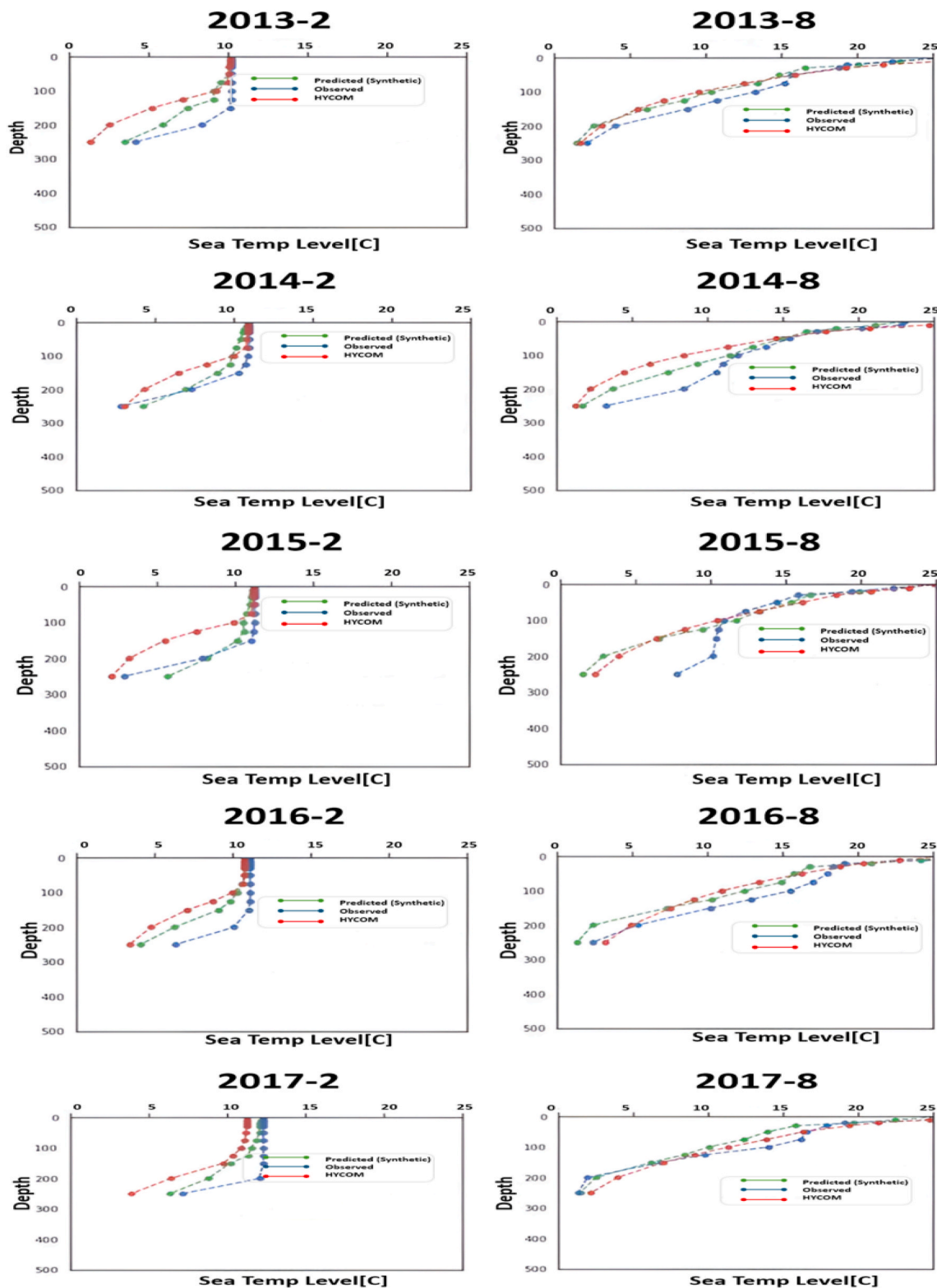


Fig. 13. Comparison of the predicted temperature profile using the stacking ensemble prediction model with the observation station (red point in Fig. 1) in February and August from 2013 to 2017. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

model using synthetic data was improved compared to the model using only observations. As shown in Fig. 12, the MAE and RMSE of most of the standalone models using the synthetic datasets were higher than those using observed data only. The MAE of KNNR using the observed dataset was 1.96 and that using the synthetic dataset was 1.03 in February. The RMSE of KNNR using the observed dataset was 2.51 and

that using the synthetic dataset was 1.67 in February. The MAE of SVR using the observed dataset was 1.49 and that using the synthetic dataset was 0.98 in February. The RMSE of SVR using the observed dataset was 2.43 and that using the synthetic dataset was 1.70. Further, in August, the MAE of KNNR using the observed dataset was 3.10 and that using the synthetic dataset was 1.73. The RMSE of KNNR using the observed

dataset was 3.60 and that using the synthetic dataset was 2.26. The MAE of SVR using the observed dataset was 2.89 and that using the synthetic dataset was 1.62. The RMSE of SVR using the observed dataset was 3.49 and that using the synthetic dataset was 2.12.

Considering the overall prediction performance in February and August, the ensemble model was better than the standalone models (Fig. 13 and Fig. S1–S3).

5. Discussion

Our study demonstrates the significant potential of using synthetic data and ensemble learning techniques to enhance subsurface temperature predictions in the EJS. The results reveal several key insights and implications for oceanographic modeling and data augmentation strategies.

5.1. Synthetic data quality and impact

The synthetic datasets generated by the generative models showed remarkable fidelity to the observed data distributions, particularly in capturing seasonal variations (Figs. 7 and 8). This high-quality synthetic data led to substantial improvements in model performance: in February, the MAE for KNNR decreased from 1.96 to 1.03 (47% improvement) when using synthetic data, and in August, the RMSE for SVR reduced from 3.49 to 2.12 (39% improvement). These results mean that the synthetic profiles did more than simply “add noise” to the training set. Instead, they filled in the gaps in our observations, especially for situations that occur rarely but are very important for the physics, such as strong Ulleung Warm Eddy events, unusually deep winter mixed layers, or very sharp summer thermoclines.

In February, the additional synthetic profiles give KNNR many more examples of deep, nearly uniform temperature structures, so the model can recognize winter mixed layer conditions more reliably and reduce its typical error by almost half. In August, the synthetic data provide SVR with a richer variety of thermocline depths and gradients, allowing it to better learn how subsurface temperature changes with SST and SSH under different eddy and stratification conditions, which leads to the 39% reduction in RMSE. In practical terms, this shows that generative models can help compensate for the limited and uneven sampling of in situ observations by creating realistic additional cases, and that this directly translates into better subsurface temperature predictions in a data sparse oceanographic setting.

5.2. Seasonal variability and model performance

The stacking ensemble model demonstrated robust performance across different seasons, accurately capturing both the winter homogeneous profiles and summer stratification (Fig. 13). This seasonal adaptability is particularly noteworthy. Winter predictions accurately represented the UWE's homogeneous water column from surface to about 200 m depth and summer predictions captured the strong thermocline, except for the anomalous case in August 2015.

The seasonal subsurface temperature structures captured by the model are physically consistent with the established dynamics of the Ulleung Warm Eddy (UWE), a dominant anticyclonic mesoscale feature in the East/Japan Sea. Previous studies have shown that the UWE strongly modulates regional circulation and sea surface height variability through the accumulation of warm, high-heat-content water in the eddy core (Mitchell, 2005; Choi et al., 2004; Lee, 2024). The model's ability to reproduce homogeneous winter profiles and strong summer thermoclines in association with SSH variability suggests that it captures physically meaningful linkages between surface geostrophic adjustment and subsurface thermal structure, rather than relying on purely statistical correlations. In addition, the UWE has been reported to play an important role in regulating nutrient redistribution and plankton productivity in the Ulleung Basin (Hyun et al., 2009; Kim et al., 2012),

indicating that the reproduced subsurface thermal patterns are also relevant to regional biogeochemical processes.

The model's ability to adapt to these seasonal changes suggests that the synthetic data effectively represented the complex oceanographic processes in the EJS. However, the slight overestimation of subsurface temperatures in August 2015 highlights the ongoing challenge of modeling transitional or anomalous events.

5.3. Ensemble learning effectiveness

The stacking ensemble outperformed the individual models and even the HYCOM reanalysis data, demonstrating the power of combining diverse learning approaches. The ensemble achieved MAE of 0.96 °C in February and 1.92 °C in August, with high correlation coefficients ($r > 0.93$). This performance surpassed that of standalone models and aligned more closely with observations than HYCOM reanalysis.

These results underscore the value of ensemble methods in oceanographic modeling, where different algorithms can capture various aspects of the complex ocean dynamics. The linear meta-learner effectively combined the strengths of KNNR (local patterns), SVR (global trends), and RFR (nonlinear interactions) to produce more accurate predictions.

5.4. Implications for oceanographic research

Our findings have several important implications for the field. First, the success of our synthetic data approach offers a promising solution to the chronic problem of data scarcity in oceanography. This method could be extended to other ocean basins or variables (e.g., salinity, currents) where observational data are limited. Second, the enhanced accuracy of subsurface temperature predictions, particularly in capturing seasonal variations and eddy dynamics, could significantly improve our understanding of heat transport and ecosystem dynamics in the EJS. Third, the model's ability to outperform HYCOM reanalysis suggests potential for developing more accurate regional ocean models, which could enhance climate predictions and marine resource management. And there is some computational efficiency. While generating synthetic data requires initial computational investment, the resulting ensemble model offers rapid operational forecasting capabilities, balancing accuracy and efficiency.

5.5. Limitations and future directions

Despite the promising results, several limitations and areas for future research emerge. First, there is anomaly detection. The model's struggle with the August 2015 anomaly suggests a need for improved representation of extreme or transitional events in the synthetic data. Second, future work should focus on ensuring that synthetic data maintains physical consistency across multiple variables (e.g., temperature-salinity relationships). And future works also need to extend the synthetic data generation and prediction to longer time scales could provide insights into decadal variability and climate change impacts. Our works also suggest multi-variable prediction. Incorporating additional oceanographic variables (e.g., salinity, currents) into the synthetic data generation and prediction model could provide a more comprehensive understanding of ocean dynamics.

In this study, the focus was on demonstrating a two stage framework in which (i) generative models first augment sparse subsurface temperature observations and (ii) the resulting synthetic-observed dataset is then used to improve the performance of several representative prediction models (KNNR, SVR, RFR, and their stacking ensemble). The framework itself is model agnostic and, in principle, can be combined with a wide range of base learners. A natural next step will be to systematically apply the same augmentation-prediction strategy to simpler baselines such as multivariate linear regression and shallow/deep artificial neural networks, and to compare their performance against the

models used here. Such an extended benchmark would more fully quantify the robustness and added value of the proposed framework across different model families and help identify the most appropriate model choices for specific operational applications.

Despite these limitations, our study demonstrates a powerful approach to enhancing subsurface temperature predictions through synthetic data generation and ensemble learning. This methodology not only improves model accuracy but also offers a scalable solution to data scarcity issues in oceanography. As we continue to refine these techniques, we anticipate significant advancements in our ability to model and understand complex ocean systems, with far-reaching implications for climate science, marine ecology, and ocean resource management.

6. Conclusion

In this study, the augmentation architecture was successfully adopted as a generative model for the subsurface temperature profile data in a marginal sea. The GAN can also be a suitable method for tabular and non-Gaussian data distribution datasets. To train a model that predicts the subsurface temperature profile in the marginal sea, the observed dataset from 1993 to 2012 was used to augment and train the data. The observed dataset from 2013 to 2017 was used to evaluate the performance of the prediction model. The augmentation architecture produced a synthetic dataset with a data distribution similar to the subsurface profile datasets. The accuracy metrics of the prediction model, the MAE was 0.96 and 1.92 and the RMSE was 1.20 and 2.45 in February and August, respectively. The performance of the model using synthetic data was improved compared to the model using only observations.

The GAN-based architecture improved and increased the real dataset for the model prediction accuracy, and a candidate served as a data imputation solution for missing values. The CopulaGAN model, which considers the correlation of variables, and TVAE are suitable for subsurface profile data synthesis.

A stacking ensemble method that combines heterogeneous models with excellent performance in the respective areas achieves a better predictive performance than a standalone model. The MAE and RMSE of the stacking ensemble were better than those of the individual regression models. To consider the characteristics of spatiotemporal distribution, based on the observation time and station points, datasets were created and trained according to the data distribution of each observed data point for better prediction. In contrast to the previous prediction model applied to the open ocean, this study can be useful in accurately predicting subsurface temperature profiles in a marginal sea with large spatiotemporal variability in water temperature owing to complex phenomena. When predicting the vertical temperature profile during the strong stratification season, it is crucial to create a predictive model that considers a thin surface mixed layer that is frequently overlooked.

The augmentation architecture was successfully adopted as a generative model for the subsurface temperature profile data in a marginal sea. Unlike conventional data augmentation techniques, our approach is specifically designed to address the challenges associated with sparse and irregular subsurface observations, ensuring the physical consistency of the generated profiles. Through comparative analysis, we demonstrated that our method enhances the diversity of training data while preserving key oceanographic structures, leading to improved predictive performance compared to traditional interpolation and generative approaches. These findings highlight the potential of the proposed method in advancing machine learning applications for oceanographic data analysis.

This study devised a method to synthesize data needed to effectively make data-based prediction models for regions with limited observations. A major achievement of this study is the use of machine learning techniques to predict subsurface data that are difficult to measure from satellite data.

This paper is based on the author's dissertation carried out at Seoul

National University (Jung, 2022).

CRediT authorship contribution statement

Kwangwoog Jung: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Yang-Ki Cho:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization. **Myeong-Taek Kwak:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that this study was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.acags.2026.100341>.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request and its original/raw dataset are available in the Korea Oceanographic Data Center, https://www.nifs.go.kr/kodc/eng/eng_coo_list.kodc.

References

- Ait-Amir, B., Pougnet, P., Hami, A.E., 2015. 6-Meta-model development. In: Hami, A.E., Pougnet, P. (Eds.), *Embedded Mechatronic Systems 2*. Elsevier, pp. 151–179. <https://doi.org/10.1016/B978-1-78548-014-0.50006-2>.
- Ali, N., Neagu, D., Trundle, P., 2019. Evaluation of k-nearest neighbour classifier performance for heterogeneous data sets. *SN Appl. Sci.* 1, 112.
- Atteia, G.E., Mengash, H.A., Samee, A., 2021. Evaluation of using parametric and non-parametric machine learning algorithms for COVID-19 forecasting. *Int. J. Adv. Comput. Sci. Appl.* 12 (10). <https://doi.org/10.14569/IJACSA.2021.0121071>.
- Auret, L., Aldrich, C., 2012. Interpretation of nonlinear relationships between process variables by use of random forests. *Miner. Eng.* 35, 27–42. <https://doi.org/10.1016/j.mineng.2012.05.008>.
- Balogun, A., Adebisi, N., 2021. Sea level prediction using ARIMA, SVR and LSTM neural network: assessing the impact of ensemble ocean-atmospheric processes on models' accuracy. *Geomat. Nat. Hazards Risk* 12 (1), 653–674. <https://doi.org/10.1080/19475705.2021.1887372>.
- Balogun, A., Rezaie, F., Pham, Q.B., Gigović, L., Drobniak, S., Aina, Y.A., Panahi, M., et al., 2021. Spatial prediction of landslide susceptibility using hybrid support vector regression models. *Geosci. Front.* 12 (3). <https://doi.org/10.1016/j.gsf.2020.10.009>.
- Brereton, R.G., Lloyd, G.R., 2010. Support vector machines for classification and regression. *Analyst* 135 (2), 230–267. <https://doi.org/10.1039/B918972F>.
- Brownlee, J., 2021. Ensemble Learning Algorithms with Python. Machine Learning Mastery. Available at: <https://machinelearningmastery.com>. (Accessed 15 March 2022).
- Burges, C.J.C., 1998. A tutorial on support vector machines for pattern recognition. *Data Min. Knowl. Discov.* 2, 121–167. <https://doi.org/10.1023/A:1009715923555>.
- Chai, T., Draxler, R.R., 2014. Root mean square error (RMSE) or mean absolute error (MAE)? *Geosci. Model Dev. (GMD)* 7, 1247–1250. <https://doi.org/10.5194/gmd-7-1247-2014>.
- Cho, Y.K., Kim, K., 1996. Seasonal variation of the East Korea warm current and its relation with the cold water. *La Mer* 34, 172–182.
- Cho, Y.K., Kim, K., 1998. Structure of the Korea strait bottom cold water and its seasonal variation. *Cont. Shelf Res.* 18 (7), 791–804.
- Cho, Y.K., Kim, K., 2000. Branching mechanism of the Tsushima current in the Korea strait. *J. Phys. Oceanogr.* 30 (11), 2788–2797.
- Choi, B.J., Haidvogel, D.B., Cho, Y.K., 2004. Nonseasonal sea level variations in the Japan/East Sea from satellite altimeter data. *J. Geophys. Res., Oceans* 109, C12028. <https://doi.org/10.1029/2004JC002387>.

- Chu, P.C., Lan, J., Fan, C., 2001. Japan Sea thermohaline structure and circulation. Part I: climatology. *J. Phys. Oceanogr.* 31 (1), 244–271. [https://doi.org/10.1175/1520-0485\(2001\)031<0244:JSTASAC>2.0.CO;2](https://doi.org/10.1175/1520-0485(2001)031<0244:JSTASAC>2.0.CO;2).
- Copernicus Climate Change Service (C3S), 2022. Sea level daily gridded data from satellite observations (1993–present). Climate Data Store. Available at: <https://cds.climate.copernicus.eu>. (Accessed 15 March 2022).
- Dubey, V., Vidyarthi, V.K., 2025. Data-driven techniques in rainfall forecasting using CMIP6 simulation outputs and ground-observed data. *Acta Geophys.* 73, 4981–4998. <https://doi.org/10.1007/s11600-025-01626-1>.
- Feng, M., Ai, W., Chen, G., Lu, W., Ma, S., 2020. A multiple linear regression algorithm for sea surface temperature retrieval. *J. Atmos. Ocean. Technol.* 37 (9), 1753–1761. <https://doi.org/10.1175/JTECH-D-20-0003.1>.
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y., 2014. Generative adversarial nets. *Adv. Neural Inf. Process. Syst.* 27.
- Gou, Y., Liu, J., Zhang, T., 2018. KNN regression model-based refinement of thermohaline data. In: *Proceedings of the 13th ACM International Conference on Underwater Networks & Systems*, pp. 1–8. <https://doi.org/10.1145/3291940.3291967>.
- Gray, P.C., Boss, E., Prochaska, J.X., Kerner, H., Begouen-Demeaux, C., Lehahn, Y., 2024. The promise and pitfalls of machine learning in ocean remote sensing. *Oceanography (Wash. D. C.)* 37 (3), 52–63. <https://doi.org/10.5670/oceanog.2024.511>.
- Gregor, L., Kok, S., Monteiro, P.M.S., 2017. Empirical methods for the estimation of Southern Ocean CO₂: support vector and random forest regression. *Biogeosciences* 14, 5551–5569. <https://doi.org/10.5194/bg-14-5551-2017>.
- Han, M., Feng, Y., Zhao, X., Sun, C., Hong, F., Liu, C., 2019. A convolutional neural network using surface data to predict subsurface temperatures. *IEEE Access* 7, 172816–172829. <https://doi.org/10.1109/ACCESS.2019.2955957>.
- HYCOM Consortium, 2022. Global ocean data assimilation experiment (1993–present). Available at: <https://www.hycom.org>. (Accessed 5 July 2022).
- Hyun, J.-H., Kim, D., Shin, C.-W., Noh, J.-H., Yang, E.-J., Mok, J.-S., Kim, S.-H., Kim, H.-C., Yoo, S., 2009. Enhanced phytoplankton and bacterioplankton production coupled to coastal upwelling and an anticyclonic eddy in the Ulleung Basin, East Sea. *Aquat. Microb. Ecol.* 54, 45–54.
- Ishfaq, H., Hoogi, A., Rubin, D., 2018. TVAE: triplet-based variational autoencoder using metric learning. *arXiv preprint*. <https://arxiv.org/abs/1802.04403>.
- Isoda, Y., Saitoh, S.-I., 1993. The northward intruding eddy along the coast of Korea. *J. Oceanogr.* 49, 443–458.
- Jeevanandam, N., 2021. What does machine learning embedding mean? *Analytics India Magazine*. Available at: <https://analyticsindiamag.com>. (Accessed 15 March 2022).
- Jiang, Y., Gou, Y., Zhang, T., Wang, K., Hu, C., 2017. A machine learning approach to argo data analysis in a thermocline. *Sensors* 17 (10), 2225. <https://doi.org/10.3390/s17102225>.
- Jiang, Y., Zhang, T., Gou, Y., He, L., Bai, H., Hu, C., 2018. High-resolution temperature and salinity model analysis using support vector regression. *J. Ambient Intell. Hum. Comput.* <https://doi.org/10.1007/s12652-018-0896-y>.
- Joh, Y., Lee, S., Park, Y.G., et al., 2024. Predictability and prediction skill of summertime East/Japan Sea surface temperature events. *npj Clim. Atmos. Sci.* 7, 210. <https://doi.org/10.1038/s41612-024-00754-7>.
- Jung, K., 2022. A Study on the Earth Science Data Generation by Numerical Modeling and Machine Learning on Cloud Computing. PhD dissertation. Seoul National University.
- Kadkhodaei, H.R., Moghadam, A.M., Dehghan, M., 2020. Heterogeneous ensemble classifier based on the boosting method and entropy measurement. *Expert Syst. Appl.* 157, 113482. <https://doi.org/10.1016/j.eswa.2020.113482>.
- Kalule, R., Abderrahmane, H.A., Alameri, W., et al., 2023. Stacked ensemble machine learning for porosity and absolute permeability prediction of carbonate rock plugs. *Sci. Rep.* 13, 9855. <https://doi.org/10.1038/s41598-023-36096-2>.
- Kang, H.E., Kang, Y.Q., 1990. Spatio-temporal characteristics of the Ulleung warm lens. *Bull. Korean Fish. Soc.* 23, 407–415.
- Kato, O., 1994. Structure of the Tsushima current in the Southwestern Japan Sea. *J. Oceanogr.* 50, 317–338.
- Kim, D., Yang, E.J., Kim, K.H., Shin, C.-W., Park, J., Yoo, S., Hyun, J.-H., 2012. Impact of an anticyclonic eddy on the summer nutrient and chlorophyll-a distributions in the Ulleung Basin, East Sea (Japan Sea). *ICES (Int. Coun. Explor. Sea) J. Mar. Sci.* 69 (1), 23–29. <https://doi.org/10.1093/icesjms/fsr178>.
- Kim, K., Kim, K.R., Chung, J., Yoo, H., Park, S., 1991. Characteristics of physical properties in the Ulleung Basin. *J. Oceanol. Soc. Korea* 26, 83–100.
- Kim, Y.-Y., Cho, Y.-K., Kim, Y.H., 2018. Role of cold water and beta-effect in the formation of the East Korean warm current in the East/Japan Sea. *Ocean Dyn.* 68 (8), 1013–1023. <https://doi.org/10.1007/s10236-018-1175-3>.
- Krider, L.A., Magner, J.A., Perry, J., Vondracek, B., Ferrington, L.C., 2013. Air–water temperature relationships in trout streams. *J. Am. Water Resour. Assoc.* 49, 896–907. <https://doi.org/10.1111/jawr.12046>.
- Langr, J., Bok, V., 2019. GANs in Action. Manning Publications, Shelter Island, NY.
- Lee, K.-J., 2024. Eddy-driven sea-level rise near the frontal region off the east coast of the Korean peninsula during 1993–2020. *Front. Mar. Sci.* 11, 1283076. <https://doi.org/10.3389/fmars.2024.1283076>.
- Leuliette, E.W., Wahr, J.M., 1999. Coupled pattern analysis of sea surface temperature and TOPEX/poseidon sea surface height. *J. Phys. Oceanogr.* 29 (4), 599–611.
- Li, Q.-J., Zhao, Y., Liao, H.-L., Li, J.-K., 2017. Effective forecast of Northeast Pacific sea surface temperature. *Atmos. Ocean. Sci. Lett.* 10 (3), 261–267. <https://doi.org/10.1080/16742834.2017.1305867>.
- Liu, J., Gou, Y., Zhang, T., Jiang, X., Du, X., Zhang, X., 2019. A-KNN: an adaptive method for constructing high-resolution ocean models. In: *2019 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*. IEEE, pp. 1–4. <https://doi.org/10.1109/ICSIDP47821.2019.9173473>.
- Loupe, G., 2014. Understanding Random Forests: from Theory to Practice. PhD dissertation. University of Liège. <https://doi.org/10.13140/2.1.1570.5928>.
- Mirza, M., Osindero, S., 2014. Conditional generative adversarial nets. *arXiv preprint*. <https://arxiv.org/abs/1411.1784>.
- Mitchell, D.A., 2005. Upper circulation patterns in the Ulleung Basin. *Deep Sea Res. Part II Top. Stud. Oceanogr.* 52, 1617–1638.
- Morrill, J.C., Bales, R.C., Conklin, M.H., 2005. Estimating stream temperature from air temperature. *J. Environ. Eng.* 131, 139–147. [https://doi.org/10.1061/\(ASCE\)0733-9372\(2005\)131:1\(139](https://doi.org/10.1061/(ASCE)0733-9372(2005)131:1(139).
- NIFS, 2022. Serial oceanographic observation data. Korea oceanographic data center. Available at: <https://www.nifs.go.kr/kodc>. (Accessed 15 March 2022).
- NOAA, 2022. High-resolution sea surface temperature analysis products. Natl. Cent. Environ. Inf. Available at: <https://www.ncei.noaa.gov>. (Accessed 15 March 2022).
- Pourzangbar, A., Jalali, M., Brocchini, M., 2023. Machine learning application in modelling marine and coastal phenomena: a critical review. *Front. Environ. Eng.* 2, 1235557. <https://doi.org/10.3389/fenv.2023.1235557>.
- Preller, R.H., Hogan, P.J., 1998. Oceanography of the Sea of Okhotsk and the Japan/East Sea. In: Robinson, A., Brink, K. (Eds.), *The Sea*, vol. 11. Wiley, pp. 429–481.
- Rocca, J., 2019. Ensemble methods: bagging, boosting and stacking. *Data Sci.* Available at: <https://towardsdatascience.com>. (Accessed 15 March 2022).
- Roemmich, D., Alford, M.H., Claustre, H., et al., 2019. On the future of argo: a global, full-depth, multi-disciplinary array. *Front. Mar. Sci.* 6, 439. <https://doi.org/10.3389/fmars.2019.00439>.
- Schmidt, J.O., Bograd, S.J., Arrizabalaga, H., et al., 2019. Future ocean observations. *Front. Mar. Sci.* 6, 550. <https://doi.org/10.3389/fmars.2019.00550>.
- Scikit-learn Developers, 2022. Scikit-learn: machine learning in python. Available at: <https://scikit-learn.org>. (Accessed 15 March 2022).
- Seo, G.-H., Cho, Y.-K., Choi, B.-J., Kim, K.-Y., Kim, B.-G., Tak, Y.-J., 2014. Climate change projection in the Northwest Pacific marginal seas. *J. Geophys. Res., Oceans* 119, 3497–3516. <https://doi.org/10.1002/2013JC009646>.
- Su, H., Wu, X., Yan, X.-H., Kidwell, A., 2015. Estimation of subsurface temperature anomaly. *Rem. Sens. Environ.* 160, 63–71. <https://doi.org/10.1016/j.rse.2015.01.001>.
- Synthetic Data Vault (SDV), 2022. Available at: <https://sdv.dev/SDV>. (Accessed 15 March 2022).
- Vidyarthi, V.K., Jain, A., Chourasiya, S., 2020. Modeling rainfall–runoff process using artificial neural network with emphasis on parameter sensitivity. *Model. Earth Syst. Environ.* 6, 2177–2188. <https://doi.org/10.1007/s40808-020-00833-7>.
- Wang, A.X., Chukova, S.S., Nguyen, B.P., 2024. Challenges and opportunities of generative models on tabular data. *Appl. Soft Comput.* 166, 111985. <https://doi.org/10.1016/j.asoc.2024.111985>.
- Wang, H., Song, T., Zhu, S., Yang, S., Feng, L., 2021. Subsurface temperature estimation from sea surface data. *Mathematics* 9, 852. <https://doi.org/10.3390/math9080852>.
- Wang, J., Song, Y., Leung, T., Rosenberg, C., Wang, J., Philbin, J., Chen, B., Wu, Y., 2014. Learning fine-grained image similarity. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1386–1393.
- Xu, L., Skoularidou, M., Cuesta-Infante, A., Veeramachaneni, K., 2019. Modeling tabular data using conditional GAN. *Adv. Neural Inf. Process. Syst.* 32.
- Yadav, P., Gaur, M., Madhukar, R.K., Verma, G., Kumar, P., Fatima, N., Sarwar, S., Dwivedi, Y.R., 2024. Rigorous experimental analysis of tabular data generated using TVAE and CTGAN. *Int. J. Adv. Comput. Sci. Appl.* 15 (4), 1250–1262. <https://doi.org/10.14569/IJACSA.2024.01504125>.
- Yao, Z., Ruzzo, W., 2006. A regression-based K nearest neighbor algorithm for gene function prediction. *BMC Bioinf.* 7, 1–11.
- Zhang, C., Ma, Y., 2012. Ensemble machine learning: methods and applications. Springer.